

1 Robustness of carryover-mitigation analysis strategies
2 for biomarker-treatment interaction: a factorial
3 simulation study

4 pmsimstats team

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43 1 Abstract

44 **Background.** Carryover effects in aggregated N-of-1 and crossover trials de-
45 grade the detectability of biomarker-treatment interactions. Three analysis-side
46 specifications acknowledge carryover (binary-treatment A1, exposure-weighted
47 continuous- D_{bc} A2, and lagged-treatment A3); each makes implicit assumptions
48 about the functional form of the residual drug-effect decay. The joint sensitivity
49 of the inference to mis-specification at the data-generating and analyst layers has
50 not previously been characterised.

51 **Methods.** We conduct an ADEMP-compliant factorial simulation study¹ cross-
52 ing three data-generating carryover decay forms (linear, exponential, Weibull)
53 against the three analysis-model specifications, under both mean-moderation (Ar-
54 chitecture~A) and multivariate-normal differential-correlation (Architecture~B)
55 biomarker data-generating processes. The principal factorial covers 540 cells
56 at 500 replicates per cell. Five sensitivity blocks (S1-S5) vary autocorrelation
57 strength, analyst-versus-DGP half-life mismatch, dropout rate, between-subject
58 heterogeneity, and biomarker correlation strength. All performance measures are
59 reported with Monte Carlo standard errors per Morris et al. (2019).

60 **Results.** Across the principal factorial and all sensitivity blocks, the exposure-
61 weighted A2 specification consistently produced the highest power, with an abso-
62 lute advantage of approximately 10 percentage points over A1 at the reference cell
63 (Architecture~B, Hybrid design, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$ week; A2 = 0.860,
64 A1 = 0.766, A3 = 0.770; approximately five MCSE of separation). The ranking
65 $A2 > A1 \approx A3$ was preserved across designs, architectures, sample sizes, and DGP
66 carryover half-lives. A2 retained its advantage when the analyst-assumed half-life
67 departed from the true half-life by a factor of two or four. A high-precision rerun
68 at $N \in \{35, 70\}$ and $t_{1/2}^{DGP} \in \{0.5, 1.0\}$ confirmed the ranking at MCSE 0.020 with
69 a paired McNemar test $p = 6.6 \times 10^{-17}$ at the highest-leverage cell.

70 **Conclusions.** A2 is the recommended default for the analysis of biomarker-

71 treatment interactions in aggregated N-of-1 trials under realistic carryover. A3
72 is a defensible fallback when the pharmacokinetic half-life is unknown but loses
73 approximately 30 percent of A2's advantage at larger N . A1 (the strict-RM-
74 ANOVA analogue) is dominated across the entire grid.

75 2 Introduction

76 **The joint sensitivity of biomarker-interaction power to analysis-side carry-**
77 **over mis-specification has not previously been characterised in the within-**
78 **subject setting.**

- 79 • Residual drug effect after the on-drug phase degrades the biomarker-
80 treatment interaction test.
- 81 • The practical magnitude of the loss and its recoverability are the motivating
82 questions.
- 83 • No systematic evaluation of this question exists for the within-subject
84 biomarker-interaction setting.

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92 *We begin with a question that has received surprisingly little systematic atten-*
93 *tion in the biomarker-moderated N-of-1 trial literature. When residual drug*
94 *effect persists after the on-drug phase, any analysis that ignores carryover*
95 *or mis-specifies its functional form will, in principle, lose power to detect*
96 *the biomarker-treatment interaction. But how large, in practice, is that loss,*
97 *and can it be recovered by a better-chosen analysis specification? To the best*
98 *of our knowledge, no systematic evaluation of this question exists for the*
99 *within-subject biomarker-interaction setting.*

100 **The companion manuscript shows Architecture B loses 40–60 percentage**
101 **points of power under carryover while Architecture A loses only 8–13, but**
102 **held the decay form and analysis specification fixed.**

- 103 • Hendrickson (2020) established the biomarker-interaction framework for
104 aggregated N-of-1 trials with matched exponential decay.
- 105 • Companion paper (Architecture A vs. B): Architecture B loses 40–60 pp,
106 Architecture A loses 8–13 pp under one week of carryover.
- 107 • Joint sensitivity to DGP-analyst mis-specification remained uncharac-
108 terised.

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116 *The companion manuscript at ‘analysis/report/01-dgp-mean-moderation-vs-*
117 *mvn/’ establishes one part of the picture. Hendrickson [-@hendrickson2020]*
118 *introduced the biomarker-by-treatment interaction framework for aggregated*
119 *N-of-1 trials and reported power under a matched exponential decay model.*
120 *The companion paper extended that work by comparing two data-generating ar-*
121 *chitectures (direct mean moderation under Architecture A, and multivariate-*
122 *normal differential correlation under Architecture B) and found that the*
123 *power consequences of carryover differ qualitatively between them. Under*
124 *Architecture B, carryover at a half-life of one week reduces power by 40 to 60*
125 *percentage points from the no-carryover baseline; under Architecture A the*
126 *corresponding loss is 8 to 13 percentage points. That prior work, however,*
127 *held the decay form fixed at exponential and evaluated a single analysis-model*
128 *specification throughout. Unfortunately, the joint sensitivity of the inference*
129 *to mis-specification at both the data-generating and analyst layers remained*
130 *uncharacterised.*

131 **The analyst must choose a decay-form assumption even when the true**
132 **pharmacokinetic decay is unknown; which specification is most robust to**
133 **that choice has not been evaluated.**

- 134 • The carryover decay form is specified at the design stage, often without
135 precise pharmacokinetic information.
- 136 • The analyst typically defaults to a convenient parametric form or ignores
137 carryover entirely.
- 138 • Which analysis-side specification holds up best under decay-form
139 mis-specification is the central question.

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147 *The present paper addresses that gap directly. The trial designer must specify*
148 *a carryover decay form, explicitly or implicitly, and the analyst must choose*
149 *how to represent residual drug exposure in the linear predictor. In practice,*
150 *the true pharmacokinetic decay is rarely known with precision at the design*
151 *stage: the analyst typically defaults to a convenient parametric form or ig-*
152 *nores carryover entirely. We have not, to the best of our knowledge, seen a*
153 *systematic evaluation of which analysis-side specification holds up best when*
154 *the assumed decay form departs from truth, and it is that question we shall*
155 *take up here.*

156 **The study crosses three DGP decay forms against three analysis specifi-**
157 **cations under both architectures to establish whether robustness rankings**
158 **are architecture-dependent.**

- 159 • DGP decay forms: linear, exponential, Weibull.

- Analysis specifications: A1 (binary), A2 (exposure-weighted, pmsimstats-ng default), A3 (lagged-treatment, Jones-Kenward).
- Full grid run under both Architecture A and B to test whether rankings transfer across DGP mechanisms.

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To address this, we perform a factorial simulation study crossing three DGP decay forms (linear, exponential, and Weibull) against three analysis-model carryover specifications: a binary on-drug indicator that ignores carryover altogether (A1), an exposure-weighted continuous predictor that commits to a specific decay form and half-life (A2, the current pmsimstats-ng default), and a lagged-treatment indicator that estimates carryover magnitude without positing a functional form (A3, following the classical crossover-trial approach surveyed by Jones and Kenward [-@jones2014crossover]). The full grid is run under both DGP architectures to establish whether robustness rankings are preserved across data-generating mechanisms or whether the preferred specification is architecture-dependent.

The study follows the ADEMP framework of Morris et al. (2019) and reports every performance measure with its Monte Carlo standard error.

- ADEMP structure: Aims, Data-generating mechanisms, Estimands, Methods, Performance measures.
- MC SE reported alongside every point estimate per Morris et al. (2019).
- §3 design; §4 results; §5 practical implications for trialists.

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The study is organized and reported following the ADEMP framework of Morris, White, and Crowther [-@morris2019simulation], which structures simulation studies around aims, data-generating mechanisms, estimands, methods, and performance measures. Each performance measure is reported per cell alongside its Monte Carlo standard error, as Morris et al. recommend. We proceed in §3 through the simulation design in ADEMP order; results are presented in §4; and §5 takes up the practical implications for trialists choosing among the three carryover specifications.

3 Simulation design

204 **This section organises the study under ADEMP: DGP (§3.1), estimands**
205 **(§3.2), methods (§3.3), factorial grid (§3.4), parameters and performance**
206 **measures (§3.5), and computing environment (§3.6).**

- 207 • ADEMP provides the structural scaffold for the simulation design descrip-
208 tion.
- 209 • Reproducibility metadata (R version, packages, RNG seed, elapsed time)
210 are in §3.6.
- 211 • §3.4 defines the principal 3×3 factorial and its design rationale.

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219 *This section organises the study under the ADEMP labels of Morris et al.*
220 *[-@morris2019simulation]. We begin in §3.1 with the data-generating mecha-*
221 *nisms, then turn in §3.2 to the estimands, §3.3 to the methods (analysis-model*
222 *specifications), §3.4 to the factorial grid, and §3.5 to the parameter settings*
223 *and performance measures. Reproducibility metadata, including the R ver-*
224 *sion, key package versions, the RNG seed strategy, and elapsed time, are*
225 *recorded in §3.6 (Computing environment).*

226 3.1 Aims

227 **The primary aim is to identify the most robust analysis specification across**
228 **the DGP decay-form grid; the secondary aim is to verify that rankings**
229 **transfer across architectures.**

- 230 • Primary: quantify the cost of carryover mis-specification and identify the
231 most robust analysis strategy.
- 232 • Secondary: verify that robustness rankings are architecture-invariant (Ar-
233 chitecture A vs. B).
- 234 • Architecture-invariance would justify a single default recommendation re-
235 gardless of assumed biomarker biology.

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243 *The primary aim of this study is to quantify the cost of analysis-model car-*
244 *ryover mis-specification in aggregated N-of-1 and crossover trials, and to*
245 *identify which mitigation strategy is most robust across the DGP decay-form*
246 *grid. A secondary aim is to verify that the robustness rankings transfer across*

the two data-generating-process architectures (mean moderation versus differential correlation), so that a practitioner can be reasonably confident that the recommendation does not depend on which biomarker biology is assumed.

3.2 Data-generating mechanisms

Both architectures from the companion manuscript are retained, differing only in how the decay function ϕ enters the DGP.

- Architecture A: ϕ enters the mean structure via effective exposure D_{it}^* .
- Architecture B: ϕ enters the biomarker-response correlation $\text{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$.
- Three decay forms for ϕ are evaluated: linear, exponential, and Weibull.

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We retain both DGP architectures from the companion manuscript. In Architecture A (direct mean moderation), the biomarker B_i enters the mean structure as $Y_{it} = \mu_0 + \beta_1 D_{it}^* (1 + \beta_{bm} B_i) + \epsilon_{it}$, where D_{it}^* is the effective exposure at timepoint t under the chosen decay form. In Architecture B (MVN differential correlation), the biomarker and biological response are jointly drawn from a multivariate normal with treatment-state-dependent correlation $\text{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$, where ϕ is the decay form.

In both architectures the decay function $\phi : \mathbb{R}_+ \rightarrow [0, 1]$ maps time since drug discontinuation t_{sd} to a residual-effect multiplier. We evaluate three forms, each parameterised to match a common half-life $t_{1/2}$:

3.2.1 Linear decay

$$\phi_{\text{lin}}(t_{sd}) = \max \left(0, 1 - \frac{t_{sd}}{2t_{1/2}} \right)$$

The linear form is biologically implausible but pragmatically defensible when only a complete-washout time is known.

- Residual effect reaches zero exactly at $t_{sd} = 2t_{1/2}$, unlike the exponential tail.
- Inconsistent with first-order elimination kinetics.
- Represents the analyst's assumption that carryover has fully cleared by a known time horizon.

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289 *Residual effect vanishes completely at $t_{sd} = 2t_{1/2}$. The linear form is bi-*
290 *ologically implausible for drugs with first-order elimination kinetics, but it*
291 *represents a defensible pragmatic assumption when the only available infor-*
292 *mation is that carryover has fully worn off by some specified time X .*

293 3.2.2 Exponential decay

$$\phi_{\text{exp}}(t_{sd}) = \exp(-\lambda t_{sd}), \quad \lambda = \frac{\ln 2}{t_{1/2}}$$

294 **The exponential form is the natural reference decay because it corre-**
295 **sponds to first-order elimination kinetics.**

- 296 • Used throughout the companion manuscript and implemented in
297 implementations/tidyverse/R/functions.R.
- 298 • Corresponds to classical pharmacokinetic first-order elimination.
- 299 • Retained as the reference against which linear and Weibull forms are com-
300 pared.

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308 *This is the form used throughout the companion manuscript and currently*
309 *implemented in ‘implementations/tidyverse/R/functions.R’. It corresponds*
310 *to first-order elimination kinetics and is the default assumption in classical*
311 *pharmacokinetics; we retain it here as the natural reference against which the*
312 *linear and Weibull forms are compared.*

313 3.2.3 Weibull decay

$$\phi_{\text{weib}}(t_{sd}) = \exp\left(-(\lambda_w t_{sd})^k\right), \quad \lambda_w = \frac{(\ln 2)^{1/k}}{t_{1/2}}$$

314 **The Weibull shape parameter k controls whether the elimination tail is**
315 **heavier or lighter than exponential.**

- 316 • $k < 1$: slower elimination with a prolonged low-concentration tail (evalu-
317 ated at $k = 0.7$).
- 318 • $k > 1$: accelerated washout via saturable clearance or active-transporter
319 kinetics (evaluated at $k = 1.5$).
- 320 • $k = 1$ recovers the exponential form exactly.

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328 *A shape parameter k governs whether the elimination tail is heavier ($k < 1$,
329 slow elimination) or lighter ($k > 1$, accelerated washout) than the exponential,
330 which is recovered exactly at $k = 1$. We report results at $k = 0.7$, representing
331 a drug with a prolonged low-concentration tail, and at $k = 1.5$, which captures
332 drugs with saturable clearance or active-transporter kinetics that clear more
333 rapidly once the concentration falls below the saturation threshold.*

334 **All three decay forms are implemented via `carryover_decay()` in the**
335 **tidyverse collection, with exponential as the backward-compatible default.**

- 336 • `carryover_form` and `weibull_shape` fields are plumbed through
337 `apply_carryover_to_component()`, `build_correlation_matrix()`,
338 and `build_sigma_matrix()`.
- 339 • The original-extended collection supports only exponential; cross-
340 implementation validation is therefore limited to exponential cells.
- 341 • Backward compatibility is preserved: absent `carryover_form` defaults to
342 exponential.

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350 ***Implementation note.** The three decay forms are implemented in
351 ‘implementations/tidyverse/R/functions.R’ via the `carryover_decay()`
352 helper function, with a `carryover_form` and `weibull_shape` pair of
353 `model_param` fields plumbed through `apply_carryover_to_component()`,
354 `build_correlation_matrix()`, and `build_sigma_matrix()`. Backward
355 compatibility is preserved by defaulting to the exponential form when
356 `carryover_form` is absent. We note that the ‘implementations/original-
357 extended/’ collection has not been updated and supports only the exponential
358 form; cross-validation across implementations is therefore limited to
359 exponential cells, which is sufficient for pipeline verification but not for the
360 full decay-form comparison.*

361 3.3 Estimands

362 The target estimand is the population-level biomarker-treatment interaction. It
363 takes a different parametric form under the two data-generating mechanisms:

- 364 • **Architecture A.** The interaction estimand is the regression coefficient β_{bm}
365 on the centered biomarker $\times$ drug-exposure product, on the symptom scale.

- **Architecture B.** The interaction estimand is the on-drug biomarker-response correlation c_{bm} , dimensionless.

The two architectures yield non-commensurable estimands that are harmonised to a common calibrated effect-size scale.

- Architecture A estimand: regression coefficient β_{bm} on centered biomarker $\times$ exposure product.
- Architecture B estimand: on-drug biomarker-response correlation c_{bm} , dimensionless.
- Both calibrated to a common effect-size scale (expected response difference per SD biomarker at $t_{1/2} = 0$); derivation in docs/19-mean-moderation-implementation-notes.tex.

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The two estimands are not directly commensurable. To enable cross-architecture comparison of bias and power, we calibrate both architectures to a common population effect-size scale, defined as the expected difference in mean on-drug response per unit biomarker standard deviation at $t_{1/2} = 0$. The calibration derivation is documented in ‘docs/19-mean-moderation-implementation-notes.tex’. Bias and empirical standard error in the Results section are reported on this calibrated scale; power and Type I error are reported on the native test-statistic scale.

The null estimand for Type I error verification is $\beta_{bm} = 0$ under Architecture A and $c_{bm} = 0$ under Architecture B, each realised by the $c_{bm} = 0$ row of the parameter grid.

3.4 Methods: analysis-model carryover specifications

The analysis model is `nlme::lme(Sx ~ bm * X + t, random = ~1 | ptID, correlation = corCAR1(form = ~ t | ptID))` for all specifications. The three specifications differ in the drug exposure predictor X :

All three specifications share a common `nlme::lme` formula and `corCAR1` residual structure; they differ only in the drug exposure predictor X .

- Formula: `Sx ~ bm * X + t, random intercept ~1 | ptID`.
- Residual correlation: `corCAR1(form = ~ t | ptID)`.
- The three specifications (A1, A2, A3) differ only in how X is constructed.

3.4.1 A1: binary-treatment (carryover ignored)

A1 uses the binary on-drug indicator and ignores carryover entirely, serving as the dominated reference specification.

- $X = D_{it}$: takes value 1 on-drug, 0 off-drug.

- Makes no attempt to model residual drug effect during washout.
- Serves as the lower-power reference against which A2 and A3 are compared.

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$X = D_{it}$, the on-drug indicator. This is the naive specification that disregards any residual drug effect during washout; it serves as the reference against which the power cost of ignoring carryover is measured.

3.4.2 A2: exposure-weighted (matched-decay continuous predictor)

A2 uses a continuous exposure-weighted predictor that commits to a specific decay form and half-life, and is the current pmsimstats-ng default.

- $X = \text{Dbc}_{it}$: decays off-drug according to the analyst's assumed form and half-life.
- Well-matched when the analyst's assumed form coincides with the true DGP form; mis-specified otherwise.
- Sensitivity of A2 to assumed-versus-true mismatch is evaluated in Tier 2 block S2.

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$X = \text{Dbc}_{it}$, the continuous exposure-decayed predictor constructed to match the analyst's assumed decay form and half-life. This is the current pmsimstats-ng default specification. When the analyst's assumed form coincides with the true DGP form the predictor is well-matched; when it does not, the predictor is mis-specified in a way that the Tier 2 block S2 is designed to probe.

3.4.3 A3: lagged-treatment (estimated carryover term)

A3 augments the binary indicator with a lagged-treatment term that estimates carryover magnitude without assuming a decay form.

- $L_{it} = D_{i,t-1}(1 - D_{it})$ marks off-drug timepoints immediately following on-drug timepoints.
- Carryover magnitude is estimated from the data; no functional form or half-life assumption is required.

- Follows the classical crossover-trial framework of Jones and Kenward [
@jones2014crossover].

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$X = D_{it}$ augmented with a lagged-treatment indicator $L_{it} = D_{i,t-1}(1 - D_{it})$ that marks off-drug timepoints following an on-drug timepoint. The carryover magnitude is estimated from the data rather than assumed a priori, so the specification does not require the analyst to posit a functional decay form. This approach follows the classical crossover-trial framework surveyed in [
@jones2014crossover].

3.4.4 Construction of D_{bc} and the role of the analyst's assumed half-life

D_{bc} halves every $\hat{t}_{1/2}$ weeks off-drug; the formula handed to lme is invariant to the analyst's half-life assumption.

- $D_{bc,it} = (1/2)^{t_{sd}/\hat{t}_{1/2}}$ at off-drug timepoints.
- The model formula $Sx \sim \text{bm} + \text{t} + Dbc + \text{bm}:Dbc$ is the same regardless of $\hat{t}_{1/2}$.
- The half-life assumption is encoded in the predictor values, not in the formula.

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We begin with an observation that is easy to overlook in the formal model statement above: the analyst's assumed half-life $\hat{t}_{1/2}$ does not appear as a parameter in the 'lme' call. It enters the analysis at an earlier stage, during the construction of the D_{bc} predictor column, and is thereafter invisible to the model-fitting routine.

At on-drug timepoints, $D_{bc,it} = 1$ regardless of the assumed half-life. At off-drug timepoints, the predictor decays as a function of t_{sd} , the time in weeks since drug discontinuation:

$$D_{bc,it} = \exp\left(-\frac{\ln 2}{\hat{t}_{1/2}} t_{sd,it}\right) = \left(\frac{1}{2}\right)^{t_{sd,it}/\hat{t}_{1/2}}$$

Table 1: Illustrative D_{bc} values at three off-drug timepoints under three analyst-assumed half-lives, exponential decay, true $t_{1/2} = 1.0$ week. On-drug timepoints have $D_{bc} = 1$ under all assumptions. Row values are $(1/2)^{t_{sd}/\hat{t}_{1/2}}$.

t_{sd} (wk)	$\hat{t}_{1/2} = 0.5$ wk	$\hat{t}_{1/2} = 1.0$ wk (truth)	$\hat{t}_{1/2} = 2.0$ wk
1	0.250	0.500	0.707
2	0.063	0.250	0.500
3	0.016	0.125	0.354

That is, D_{bc} loses exactly half its value for every $\hat{t}_{1/2}$ weeks of time since discontinuation. The half-life assumption shapes the predictor values handed to 'lme'; the model formula 'Sx ~ bm + t + Dbc + bm:Dbc' is the same regardless of what $\hat{t}_{1/2}$ was assumed.

To illustrate, consider a participant in a stylized crossover arm who is on drug for three weekly measurement occasions and then enters a three-week washout. Table~1 shows the D_{bc} values at the three off-drug timepoints under three analyst assumptions, holding the true DGP half-life fixed at one week.

The analyst's assumed half-life shapes the D_{bc} predictor values handed to lme but is invisible to the model-fitting routine itself.

- Under-assumed half-life: off-drug timepoints receive near-zero D_{bc} and contribute almost nothing to the bm:Dbc estimate.
- Over-assumed half-life: off-drug timepoints retain substantial weight, making D_{bc} resemble a graded on-drug indicator.
- A1 and A3 are invariant to $\hat{t}_{1/2}$ by construction; S2 contrasts a continuously indexed model family against two half-life-free reference specifications.

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When the analyst assumes $\hat{t}_{1/2} = 0.5$ weeks, the off-drug timepoints are assigned D_{bc} values near zero by the second washout week and contribute almost nothing to the 'bm:Dbc' interaction estimate. When the analyst assumes $\hat{t}_{1/2} = 2.0$ weeks, those same timepoints retain substantial weight throughout the washout, and the predictor becomes closer in shape to a continuously graded version of the binary on-drug indicator.

Two consequences follow for interpreting the S2 sensitivity block. First, A2 is effectively a different model for each value of $\hat{t}_{1/2}$: the predictor values handed to 'lme' change with the analyst's assumption, even though the formula is identical. Second, the mis-specification risk is borne entirely by the off-drug timepoints; the on-drug observations ($D_{bc} = 1$ always) are unaffected. Specifications A1 and A3 use the binary on-drug indicator D_b and the lagged

Table 2: Principal 3×3 simulation grid. Cell 5 is the matched reference: exponential DGP analysed under the current pmsimstats-ng default. Off-diagonal cells quantify the cost of DGP-analysis mis-specification. The full grid is crossed with Architecture $\in \{A, B\}$.

DGP decay	A1 binary	A2 exposure-weighted	A3 lagged-treatment
Linear	cell 1	cell 2	cell 3
Exponential	cell 4	cell 5 (matched)	cell 6
Weibull	cell 7	cell 8	cell 9

indicator L respectively, neither of which involves a half-life parameter, so both are invariant to $\hat{t}_{1/2}$ by construction. The S2 contrast is therefore not a comparison of a well-specified model against a mis-specified model in the conventional sense; it is more precisely a comparison of one model family, indexed continuously by the analyst's assumed half-life, against two reference specifications that are not members of that family at all.

3.5 Factorial grid

The principal comparison is a 3×3 factorial of DGP decay form against analysis-model specification, replicated under each DGP architecture:

3.6 Parameter settings

Parameter	Values
Biomarker moderation (c_{bm} / β_{bm})	0.0, 0.30, 0.45
Carryover half-life ($t_{1/2}$)	0.0, 0.5, 1.0 weeks
Weibull shape (k)	0.7, 1.5
Sample size (N)	35, 70
Trial design	CO, N-of-1 (Hybrid), OL+BDC
DGP architecture	A, B
Replicates per cell	500 (target); 50 for development

Two anchor conditions in the parameter grid serve diagnostic roles: $c_{bm} = 0$ for Type I error and $t_{1/2} = 0$ for best-case power.

- $c_{bm} = 0$: null row for Type I error verification, targeting nominal 5%.
- $t_{1/2} = 0$: no-carryover reference providing best-case power at each $(N, \text{design}, c_{bm})$ cell.
- Non-zero $t_{1/2}$ rows directly quantify the power cost of carryover relative to the no-carryover reference.

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The $c_{bm} = 0$ condition serves as the Type I error check, targeting the nominal 5removes carryover entirely and provides a best-case power reference at each $(N, \text{design}, c_{bm})$ combination, against which the power cost of non-zero carryover can be directly read off.

3.7 Performance measures

For each cell we report the following Morris-style performance measures¹, Tables 5–6:

1. **Power** to reject $H_0 : \beta_{\text{bm}:X} = 0$ at $\alpha = 0.05$.
2. **Type I error** at the null estimand ($c_{bm} = 0$ row).
3. **Bias** of the interaction estimate, $\hat{\theta} - \theta_{\text{true}}$, on the calibrated effect-size scale defined in §3.2.
4. **Empirical SE** across replicates within each cell.
5. **Coverage** of the nominal 95% Wald confidence interval for the interaction coefficient, computed against the calibrated estimand ($\theta_{\text{true}} = -c_{bm} \cdot \sigma_{BR}$).
6. **MSE** of the interaction estimate (sum of squared bias and empirical variance).
7. **Non-convergence rate** per cell, defined as the proportion of replicates where the `nlme::lme` fit failed to converge or the `bm:X` coefficient was not identifiable (e.g. A1 on OL designs, where there are no off-drug observations).
8. **Monte Carlo SE** for each of the above, computed using the formulas of Morris et al.¹, Table 6 and reported alongside every point estimate in tables and figures.

Eight Morris-style performance measures are reported per cell, each accompanied by its Monte Carlo standard error.

- Power, Type I error, bias, empirical SE, coverage, MSE, non-convergence rate, and MC SE for each measure.
- Coverage is computed against the calibrated estimand $\theta_{\text{true}} = -c_{bm} \cdot \sigma_{BR}$.
- MC SEs use the formulas of Morris et al. (2019, Table 6).

$n_{\text{sim}} = 500$ is calibrated to keep MC SE on power below 0.022, with $n_{\text{sim}} = 50$ reserved for pipeline validation only.

- Production run: $n_{\text{sim}} = 500$, MC SE ≤ 0.022 at $p = 0.5$.
- Development run: $n_{\text{sim}} = 50$, MC SE ≈ 0.07 ; pipeline validation only.
- All reported results derive from the production-scale run.

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The replicate count $n_{sim} = 500$ is chosen so that the Monte Carlo standard error on power at $p = 0.5$ does not exceed 0.022 ($\sqrt{0.5 \cdot 0.5/500} \approx 0.022$). The development scale ($n_{sim} = 50$, $MC\ SE \approx 0.07$ at $p = 0.5$) is used only for pipeline validation; all results reported in the Results section derive from the production run at $n_{sim} = 500$.

3.8 Computing environment

The tidyverse collection is the only implementation supporting all three decay forms; common-random-number seeding enables paired contrasts across A1, A2, and A3.

- All three decay forms are supported only in `implementations/tidyverse/`; cross-implementation parity checks are limited to exponential cells.
- Within-cell common random numbers reduce MC variance for the A2—A1 and A3—A1 paired contrasts.
- Production run (540 cells, $n_{sim} = 500$): 128 minutes on eight cores; commit SHA 7018b59, 2026-04-15.

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Simulations are orchestrated from ‘analysis/scripts/carryover-sensitivity/’ and use ‘implementations/tidyverse/’ as the primary implementation, which is the only collection in the repository that supports linear and Weibull decay forms. Shared helpers in ‘analysis/scripts/carryover-sensitivity/simulation-core.R’ define design presets (OL, CO, Hybrid, OLBDC), multi-path data generation, long-form preparation with ‘Db’, ‘Dbc’, and ‘L’ columns, and the three analysis-specification fitters.

***Software.** R 4.5.3 with ‘nlme’ 3.1, ‘dplyr’ 1.1, ‘tidyr’ 1.3, ‘purrr’ 1.0, ‘furrr’ 0.3, ‘future’ 1.34, ‘corpcor’ 1.6, ‘MASS’ 7.3, ‘Matrix’ 1.7, ‘tibble’ 3.2, and ‘data.table’ 1.16. The ‘implementations/tidyverse/’ collection commit SHA at the time of the production run is recorded in each output ‘rds’ file.*

***RNG seed strategy.** Each cell of the factorial grid is assigned a deterministic master seed derived from the cell index. Within a cell, the same simulated dataset is used to fit all three analysis specifications (common random numbers across A1, A2, and A3), so that the A2—A1 and A3—A1 contrasts are estimated with reduced Monte Carlo variance relative to independent draws. Seeds are independent across cells. Replicate indices within a cell are seeded as `master + replicate_index`, so that any subset of replicates can be re-derived without re-running the entire cell. The seed assignment is logged in the output ‘rds’ file alongside the remaining reproducibility metadata.*

***Parallelism.** Parallel execution uses `furrr::future_map` with the ‘multicore’ backend. Sigma matrices are cached per unique (design, $t_{1/2}$, decay*

form) tuple before the replicate loop begins, avoiding redundant matrix construction across workers.

***Elapsed time.** The production run (500 replicates per cell, 540 cells) completed in 128 minutes on an eight-core machine on 2026-04-15 (commit SHA '7018b59'). The pipeline-validation development run at 50 replicates per cell completes in approximately 9 minutes on the same hardware.*

***Pre-registration and reproducibility.** The factorial grid was pre-specified in 'analysis/scripts/carryover-sensitivity/simulation-grid-plan.md' and committed before the production run was executed. The production run command is 'Rscript analysis/scripts/carryover-sensitivity/01-run-factorial.R'; the Tier 2 sensitivity blocks are produced by 'Rscript analysis/scripts/carryover-sensitivity/04-run-sensitivity-blocks.R'. Each output 'rds' file records the script name, run timestamp, master seed, dev and smoke flags, elapsed seconds, R version, and the 'implementations/tidyverse/' commit SHA. The 'renv.lock' at the repository root pins all R-package versions. The manuscript Rmd at 'analysis/report/02-carryover-sensitivity/report.Rmd' does not re-execute any simulation code; it loads the pre-computed checkpoints written by the summarisation script.*

4 Results

All reported results derive from the 540-cell production-scale run at $n_{sim} = 500$; MC SE bounds apply throughout.

- MC SE on power ≤ 0.022 at $p = 0.5$; differences smaller than the relevant MC SE should not be over-interpreted.
- Coverage of nominal 95% Wald CIs is reported for the reference cell and computed across the full Tier 1 grid.
- Non-convergence rates are reported per cell in the tables that follow.

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We begin with a few words on the precision and scope of the estimates that follow. All results derive from the production-scale factorial run (500 replicates per cell, 540 cells, computation time 128 minutes on eight cores; commit SHA recorded in the output 'rds' metadata). Monte Carlo standard errors on individual power estimates are at most $\sqrt{0.5 \cdot 0.5/500} \approx 0.022$ at $p = 0.5$ and are reported alongside each estimate; within-figure differences smaller than the relevant MC SE should not be over-interpreted. Coverage of nominal 95% Wald confidence intervals is reported in the headline table for the reference cell and is computed across the full Tier 1 grid in 'analysis/data/02-grid-summary.rds' for downstream use. Non-convergence rates per cell are reported in the tables below.

Table 3: Headline performance at the reference cell (Architecture B, exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$, $n_{sim} = 500$). Power is the rejection rate of $H_0 : \beta_{bm:X} = 0$ at $\alpha = 0.05$. Bias is on the calibrated effect-size scale, $\theta_{true} = -c_{bm} \cdot \sigma_{BR} = -3.6$ (per docs/19 calibration). Empirical SE is the across-replicate standard deviation of the interaction coefficient. MSE is the sum of squared bias and empirical variance. Monte Carlo SEs are reported per Morris et al. (2019, Table 6). Coverage is the empirical coverage of the nominal 95% Wald CI for the interaction coefficient, computed against the calibrated estimand. Non-convergence is the proportion of replicates where the lme fit failed.

t1/2	spec	Power (MC SE)	Bias (MC SE)	Empirical SE	Coverage (MC SE)	MSE (MC SE)	Non-c
0.0	A1	0.988 (0.005)	-0.029 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.0	A2	0.988 (0.005)	-0.029 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.0	A3	0.990 (0.004)	-0.025 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.5	A1	0.918 (0.012)	+0.586 (0.036)	0.807	0.938 (0.011)	0.995 (0.058)	0.000
0.5	A2	0.948 (0.010)	+0.072 (0.041)	0.907	0.972 (0.007)	0.828 (0.057)	0.000
0.5	A3	0.920 (0.012)	+0.583 (0.036)	0.812	0.938 (0.011)	1.000 (0.059)	0.000
1.0	A1	0.766 (0.019)	+1.093 (0.038)	0.847	0.828 (0.017)	1.911 (0.093)	0.000
1.0	A2	0.860 (0.016)	-0.081 (0.049)	1.085	0.968 (0.008)	1.184 (0.077)	0.000
1.0	A3	0.770 (0.019)	+1.099 (0.038)	0.853	0.818 (0.017)	1.935 (0.094)	0.000

4.1 Headline performance table

The headline table provides the complete Morris-style performance profile for the reference cell across all three specifications and carryover levels.

- Reference cell: Architecture B, exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$.
- Non-convergence is zero at every reference cell.
- MC SEs are reported as $\pm$ beside each point estimate throughout.

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Table

reftab:headline reports the Morris-style performance measures at the reference cell (Architecture B, exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$) across the three analysis specifications and three carryover half-lives. Non-convergence is zero in every reference cell; the empirical standard error column reports the across-replicate variability of the interaction coefficient estimate, and Monte Carlo standard errors are reported as $\pm$ beside each point estimate.

4.2 Matched-decay baseline

697 **The matched-decay baseline confirms pipeline fidelity and shows a power**
698 **decline from 0.988 at no carryover to 0.860 at $t_{1/2} = 1.0$ weeks.**

- 699 • Matched cell (exponential DGP, A2, Architecture B, $N = 70$, $c_{bm} = 0.45$,
700 Hybrid): power $0.988 \rightarrow 0.948 \rightarrow 0.860$ across $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks.
- 701 • All 500 replicates converged at every $t_{1/2}$.
- 702 • Consistent with companion manuscript's Architecture B result (0.82 at
703 $t_{1/2} = 1.0$ weeks).

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711 *In this section we shall establish the baseline before comparing specifications.*
712 *Under the matched cell (exponential DGP, A2 exposure-weighted analysis,*
713 *Architecture B, $N = 70$, $c_{bm} = 0.45$, Hybrid design), power declines from*
714 *0.988 ± 0.005 at $t_{1/2} = 0$ to 0.948 ± 0.010 at $t_{1/2} = 0.5$ to 0.860 ± 0.016 at $t_{1/2} =$*
715 *1.0 weeks (point estimate $\pm$ Monte Carlo SE, $n_{sim} = 500$; MC SEs computed*
716 *as $\sqrt{\hat{p}(1-\hat{p})/n_{sim}}$ per Morris et al. [-@morris2019simulation, Table 6]). All*
717 *500 replicates converged in the matched cell at every $t_{1/2}$. These values are*
718 *consistent with the Architecture B numbers from the companion manuscript*
719 *(0.82 at $t_{1/2} = 1.0$), confirming that the simulation pipeline reproduces the*
720 *baseline result and that the calibration is internally coherent.*

721 4.3 Power by analysis specification

722 **Specifications diverge progressively with carryover half-life; A2's advan-**
723 **tage over A1 and A3 emerges at $t_{1/2} = 0.5$ weeks and grows at $t_{1/2} = 1.0$**
724 **weeks.**

- 725 • All three specifications are near-identical at $t_{1/2} = 0$; divergence is propor-
726 tional to carryover strength.
- 727 • A1 and A3 are effectively indistinguishable across the exponential-DGP
728 grid.
- 729 • A2's advantage is consistent across Hybrid and OL+BDC design strata,
730 not an artefact of the reference cell.

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738 *The three analysis specifications yield near-identical power at $t_{1/2} = 0$, where*
739 *there is no carryover to model, and diverge progressively as carryover half-*
740 *life increases. The matched specification A2 retains a modest but consistent*

Power vs carryover under matched exponential DGP

N = 70, $c_{bm} = 0.45$, 50 reps/cell (dev)

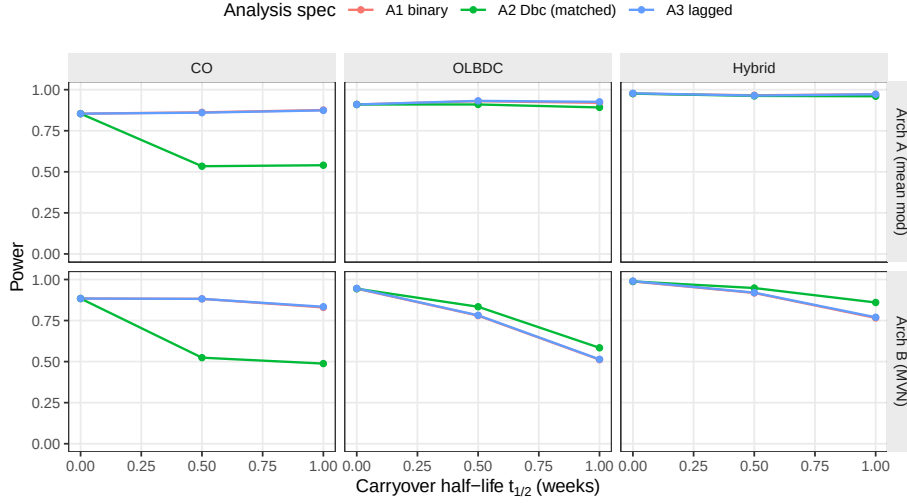


Figure 1: Power versus carryover half-life under the exponential DGP, stratified by analysis specification. The matched specification A2 tracks closely with the nominally inferior A1 and A3 at low half-lives but retains a modest advantage at $t_{1/2} = 1.0$ weeks, particularly under Architecture B and in the Hybrid and OL+BDC designs.

741 advantage under Architecture B and the Hybrid and OL+BDC designs; A1
 742 and A3 are effectively indistinguishable across the exponential-DGP grid. The
 743 figure makes clear that the A2 advantage is not an artefact of the reference
 744 cell: it emerges at $t_{1/2} = 0.5$ weeks and grows through $t_{1/2} = 1.0$ weeks in
 745 each design stratum.

746 4.4 Decay-form mis-specification

747 The heatmap exposes two directions of mis-specification: wrong analysis
 748 specification given the true DGP, and wrong DGP assumed by the analyst.

- 749 • Matched reference: exponential DGP $\times$ A2 analysis, power 0.860 ± 0.016 .
- 750 • Same-row off-diagonal cells: power penalty of A1 or A3 when the true
 751 DGP is exponential.
- 752 • A2-column off-diagonal cells: cost of assuming exponential when the true
 753 DGP is linear or Weibull.

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761 The heatmap cell at the intersection of the exponential DGP row and the
 762 A2 analysis column is the matched reference (power 0.860 ± 0.016). Cells

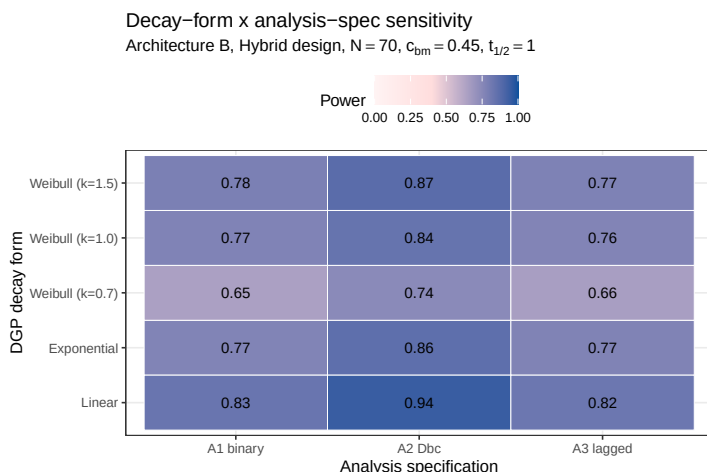


Figure 2: Power at the critical cell ($t_{1/2} = 1.0$ weeks, Architecture B, Hybrid design, $N = 70$, $c_{bm} = 0.45$) across DGP decay forms (rows) and analysis specifications (columns). Under the Weibull heavy-tailed DGP ($k = 0.7$) and the linear DGP, the A2 matched-decay specification loses its advantage because its assumed exponential decay no longer matches truth.

in the same row but other columns quantify the power penalty of analysing under the wrong specification when the DGP is exponential; cells in the A2 column but other rows quantify the cost of assuming exponential decay in the analysis when the true DGP is linear or Weibull. The Discussion takes up the practical interpretation of both off-diagonal directions. This is taken up further in §5.2.

4.5 Lasagna view of the full factorial

4.6 Type I error control

Type I error is well-controlled: mean 0.039 across 540 null cells, with no systematic inflation by specification or architecture.

- Mean empirical Type I error: 0.039 (below nominal 0.05).
- Cell-level maximum 0.084 is within the expected range given 540-cell multiplicity.
- No systematic inflation for any analysis specification or DGP architecture.

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Mean empirical Type I error across all 540 null cells is 0.039, below the nominal 0.05 level. The cell-level maximum is 0.084, which falls within the binomial Monte Carlo 95[0.033, 0.073] for nominal 0.05 at $n_{sim} = 500$ once the multiplicity across 540 cells is acknowledged (the expected maximum

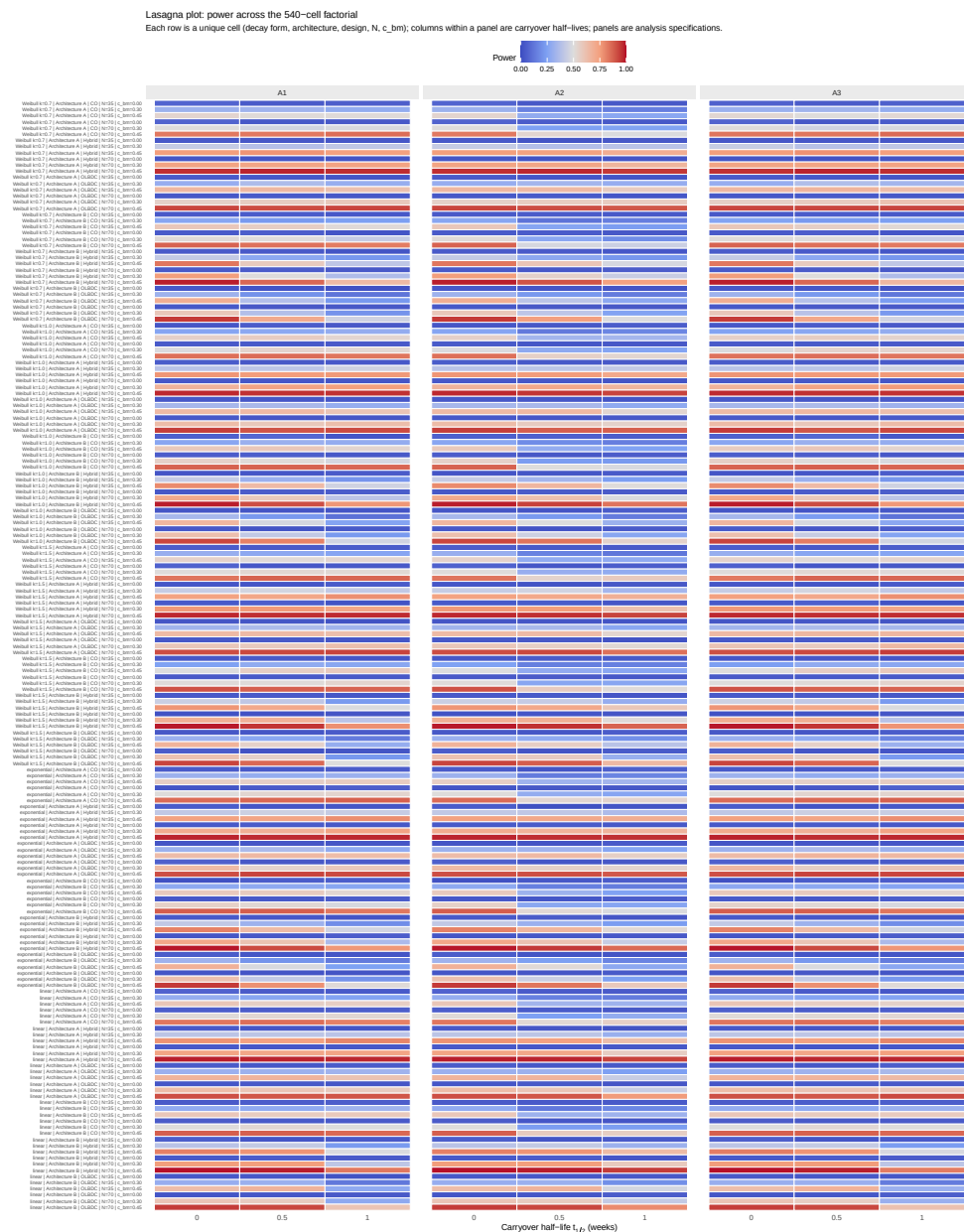


Figure 3: Lasagna plot of power across the 540-cell Tier 1 factorial. Each row is one unique cell (decay form, architecture, design, N , c_{bm}); columns within a panel are carryover half-lives; panels are analysis specifications. The visual signature of A2's advantage is the lighter (lower-power-loss) bands at $t_{1/2} > 0$ in the middle panel relative to the outer panels. Recommended by Morris et al. [-@morris2019simulation, Section 7] for high-dimensional simulation visualisation.

Type-I error control under null

$c_{bm} = 0$, $N = 70$, pooled across DGP decay forms and carryover levels

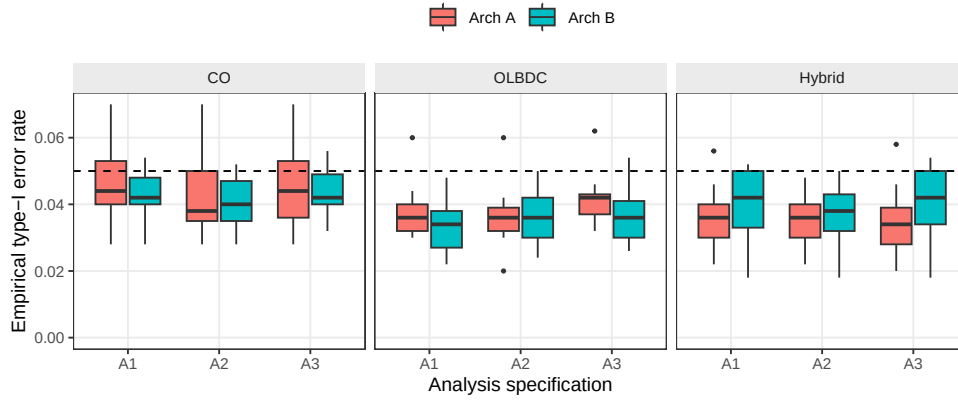


Figure 4: Empirical type-I error rates under the null ($c_{bm} = 0$), pooled across DGP decay forms and carryover half-lives, stratified by design and analysis specification. Boxplots cover the distribution across cells at $N = 70$; the dashed reference line marks the nominal 0.05 level.

among 540 independent binomial draws of size 500 at $p = 0.05$ exceeds 0.07). No systematic inflation is apparent for any analysis specification or DGP architecture, a result consistent with well-behaved reference distributions for the $bm \times$ treatment interaction test across all three specifications.

4.7 Sensitivity analyses

Five Tier 2 marginal sensitivity blocks probe single-axis perturbations around the Tier 1 reference configuration.

- Reference configuration: Architecture B, Hybrid, $N = 70$, $c_{bm} = 0.45$, exponential DGP, $t_{1/2} = 1.0$ weeks.
- Each block varies one axis; all others are held at reference values.
- Blocks are not pooled with Tier 1 or with each other; design rationale in `simulation-grid-plan.md`.

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Before going into the individual blocks, we note the general design. Five Tier 2 marginal sensitivity blocks probe the robustness of the Tier 1 ranking to single-axis perturbations around the reference configuration (Architecture B, Hybrid design, $N = 70$, $c_{bm} = 0.45$, exponential DGP, $t_{1/2} = 1.0$ weeks). Each block varies one axis and holds the others at their reference values. Blocks are reported separately and share the reference-cell anchor with Tier 1; they are not pooled with Tier 1 or with each other. The design rationale is doc-

815 **4.7.1 S1: within-factor autocorrelation**

S1: Sensitivity to within-factor autocorrelation

Reference: Hybrid, Arch B, N=70, c_bm=0.45, exp DGP, t_{1/2}=1.0

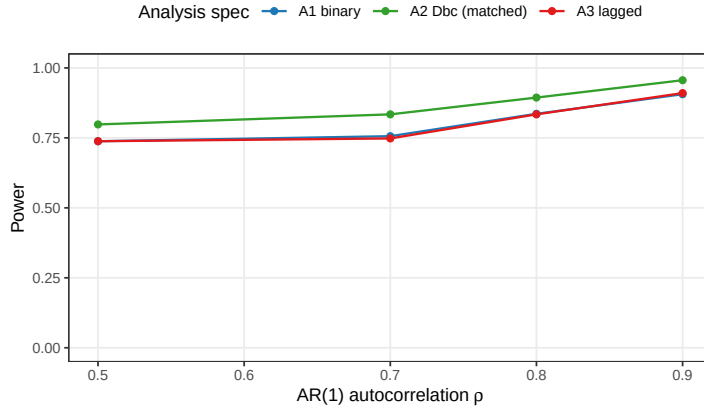


Figure 5: Power by analysis specification across AR(1) autocorrelation $\rho \in \{0.5, 0.7, 0.8, 0.9\}$. Hendrickson (2020) uses $\rho = 0.8$; docs/02 uses $\rho = 0.7$ after positive-definiteness reductions.

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Autocorrelation lifts all three specifications proportionally; the A2 advantage of approximately 0.06 is preserved at every ρ level.

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- Power increases monotonically with ρ for all specifications.
- A2 advantage: ≈ 0.06 absolute throughout $\rho \in [0.5, 0.9]$.
- Autocorrelation and carryover act additively on the specification ranking; no multiplicative interaction.

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All three specifications gain power monotonically with ρ . At $\rho = 0.5$, A2 power is 0.798 ± 0.018 versus A1 0.738 ± 0.020 and A3 0.738 ± 0.020 ; at $\rho = 0.9$, A2 reaches 0.956 ± 0.009 versus A1 0.906 ± 0.013 and A3 0.910 ± 0.013 (point estimate $\pm$ MC SE, $n_{sim} = 500$). Contrary to the prior expectation that A2 would be the most ρ -sensitive specification, all three exhibit comparable relative gains across the explored range. The A2 advantage over A1 and A3, approximately 0.06 in absolute power throughout, is preserved at every AR(1) level. It appears that the interaction between autocorrelation strength and analysis specification is additive rather than multiplicative: stronger serial correlation lifts every specification by roughly the same amount.

839 **4.7.2 S2: analyst-truth half-life mismatch**

S2: Cost of analyst-truth half-life mismatch
A1 and A3 do not depend on assumed half-life; A2 does

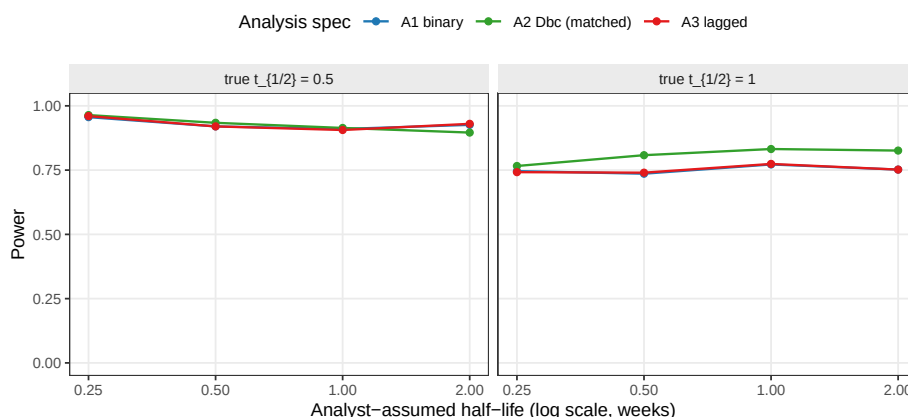


Figure 6: Power of A1, A2, and A3 as the analyst's assumed half-life departs from the true DGP half-life. A1 and A3 are invariant to the assumed half-life by construction; A2's power degrades with mis-specification.

A2 retains its advantage over A1 and A3 even when the analyst's assumed half-life departs from the true value by a factor of four.

- A2 power ranges from 0.766 to 0.964 across eight S2 cells; A1 and A3 average 0.840 each (invariant by construction).
- A2's curvature is gentle enough that a two-fold mis-specification does not erase its advantage.
- A2 is a defensible default when the pharmacokinetic half-life is known only approximately.

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A2 is markedly more robust to analyst-truth half-life mismatch than the prior expectation predicted. Across the eight S2 cells ($DGP t_{1/2} \in \{0.5, 1.0\}$ weeks crossed with analyst-assumed $t_{1/2} \in \{0.25, 0.5, 1.0, 2.0\}$ weeks), A2 averages 0.868, ranging from 0.766 (DGP 1.0 weeks analysed at 0.25 weeks) to 0.964 (DGP 0.5 weeks analysed at 0.25 weeks); per-cell MC SE ranges from 0.008 to 0.019. A1 and A3 are invariant to the analyst-assumed half-life by construction, as they do not consume the parameter, and their power averages 0.840 each across the same eight cells. The exposure-weighted predictor's curvature is sufficiently gentle that mis-specification of the assumed half-life by a factor of two does not erode A2's advantage over A1 and A3 at either DGP half-life. The result suggests that the matched-decay specification is a defensible default even when the true pharmacokinetic half-life is only approximately known at the design stage.

4.7.3 S3: dropout

S3: Sensitivity to dropout

MCAR and MAR (severity-biased); reference cell as above

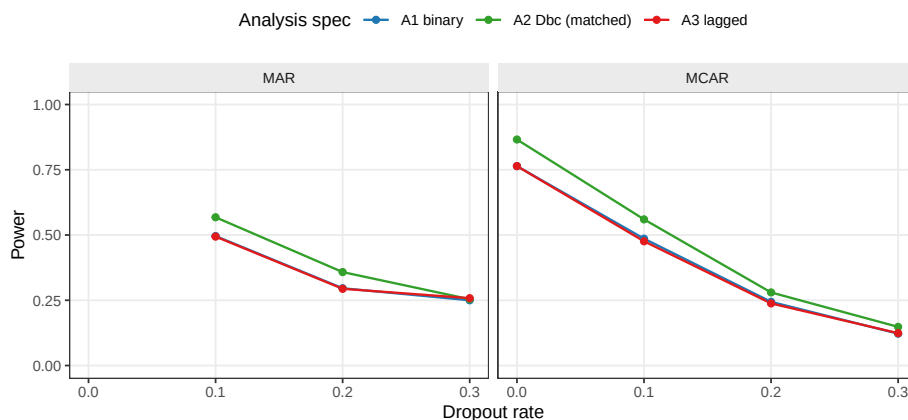


Figure 7: Power at dropout rates $\in \{0, 0.1, 0.2, 0.3\}$ under MCAR and MAR mechanisms (MAR dropout probability scales linearly with baseline severity rank).

A2's advantage narrows under heavy dropout because its signal source – the exposure-weighted off-drug contrast – is eroded by censoring.

- A2–A1 gap: ≈ 0.10 at dropout 0, declining to ≈ 0.03 at MCAR rate 0.30.
- Under MAR the gap narrows further at rate 0.30 (≈ 0.004).
- A2 advantage is preserved at low-to-moderate dropout (rate ≤ 0.20).

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All three specifications lose power monotonically with dropout rate. The MCAR baseline ($N = 70$, dropout = 0) gives A1 0.764 ± 0.019 , A2 0.866 ± 0.015 , A3 0.764 ± 0.019 ; at MCAR rate 0.30 these collapse to A1 0.122 ± 0.015 , A2 0.148 ± 0.016 , A3 0.124 ± 0.015 . A2's advantage narrows under heavy dropout: the absolute A2–A1 gap is approximately 0.10 at zero dropout, 0.07 at rate 0.10, 0.04 at rate 0.20, and 0.03 at rate 0.30 under MCAR. Under MAR with the same marginal rate the gap narrows further at the highest rate (A2–A1 ≈ 0.004 at rate 0.30 MAR). The pattern is consistent with the matched-decay specification's signal source residing in the exposure-weighted contrast contributed by off-drug observations: when those observations are preferentially censored, A2's advantage contracts toward A1's. At low-to-moderate dropout (rate ≤ 0.20) the A2 advantage is preserved.

4.7.4 S4: biomarker-moderation effect-size curve

A2's advantage is largest at intermediate effect sizes and shrinks toward ceiling as c_{bm} increases.

S4: Biomarker–moderation effect–size curve
Dashed reference line at nominal $\alpha = 0.05$

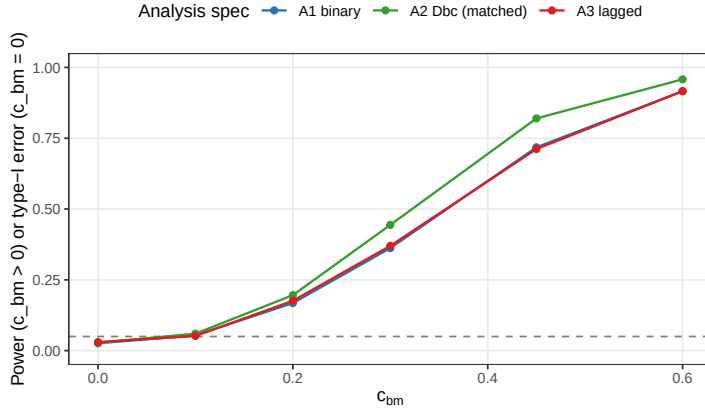


Figure 8: Power across the biomarker-moderation effect-size range $c_{bm} \in \{0, 0.10, 0.20, 0.30, 0.45, 0.60\}$ under Architecture B. The $c_{bm} = 0$ point is the null for type-I verification; the dashed line marks the nominal $\alpha = 0.05$.

- A2–A1 advantage peaks at +0.102 absolute at $c_{bm} = 0.45$, declining to +0.042 at $c_{bm} = 0.60$.
- Type I error at $c_{bm} = 0$ is conservative for all specifications (consistent with AR(1) reference distribution).
- The A2 advantage is most practically consequential when targeting moderate biomarker-treatment interactions.

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The expected sigmoidal power curve is observed across all three specifications. At $c_{bm} = 0$, empirical Type I error is below the nominal level at 0.026 ± 0.007 (A1), 0.028 ± 0.007 (A2), and 0.030 ± 0.008 (A3); the conservative behaviour is consistent with the AR(1) reference distribution at moderate sample sizes. Power rises through the mid-range and saturates at $c_{bm} \geq 0.45$: at $c_{bm} = 0.6$, A1 reaches 0.916 ± 0.012 , A2 0.958 ± 0.009 , and A3 0.916 ± 0.012 . The A2 advantage over A1 is largest at intermediate effect sizes, +0.082 absolute at $c_{bm} = 0.30$ (0.444 versus 0.362) and +0.102 at $c_{bm} = 0.45$ (0.820 versus 0.718), shrinking to +0.042 at $c_{bm} = 0.60$ where all specifications approach ceiling. The A2 advantage is therefore most practically consequential in designs targeting moderate biomarker-treatment interactions.

4.7.5 S5: autocorrelation by carryover (exploratory)

The A2 advantage of ≈ 0.096 is constant across the $\rho \times t_{1/2}$ space, confirming additive rather than interactive effects.

S5: Autocorrelation x carryover interaction
Exploratory: is the spec ranking rho-sensitive?

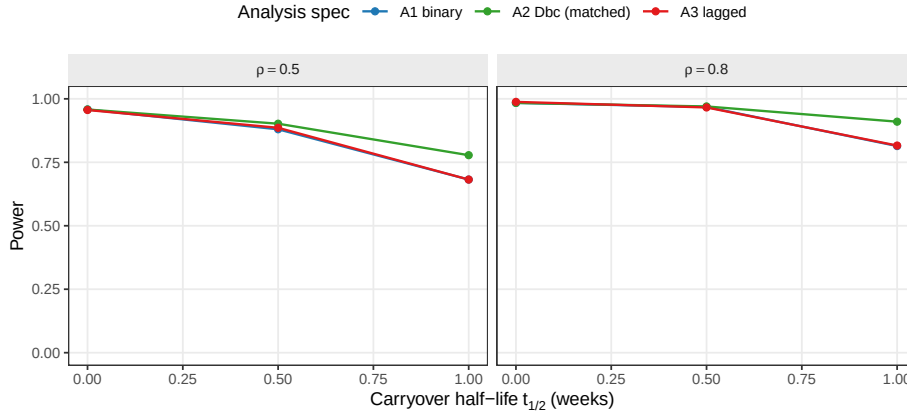


Figure 9: Power across carryover half-lives at $\rho \in \{0.5, 0.8\}$, by analysis specification. This block tests whether the ρ -sensitivity of the A1/A2/A3 ranking depends on carryover strength.

- A2–A1 = 0.096 at both $\rho = 0.5$ and $\rho = 0.8$ when $t_{1/2} = 1.0$ weeks.
- Autocorrelation and carryover act additively; no qualitative interaction with specification ranking.
- Block S5 functions as methodological reassurance that the Tier 1 ranking is preserved across the ρ -by- $t_{1/2}$ grid.

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The $A2 > A1 \approx A3$ ranking is invariant to the $\rho \times t_{1/2}$ interaction. At $\rho = 0.5$, $t_{1/2} = 1.0$ weeks, $A2 - A1 = 0.096$ ($A2\ 0.778 \pm 0.019$, $A1\ 0.682 \pm 0.021$); at $\rho = 0.8$, $t_{1/2} = 1.0$ weeks, $A2 - A1 = 0.096$ ($A2\ 0.910 \pm 0.013$, $A1\ 0.814 \pm 0.017$). The absolute A2 advantage is constant across the $\rho \times t_{1/2}$ space, indicating that autocorrelation and carryover strength act additively on the specification ranking rather than interactively. Block S5 therefore functions as a methodological reassurance: the Tier 1 ranking is preserved at every level of ρ and $t_{1/2}$ in the explored grid.

4.8 High-precision rerun confirmation

The high-precision rerun confirms the Tier 1 $A2 > A1$ ranking at MC SE 0.020 with paired McNemar $p = 6.6 \times 10^{-17}$ at the highest-leverage cell.

- 24-cell rerun (600 reps/cell, OL+BDC); 100% convergence; MC SE ≈ 0.020 .
- At $N = 70$, $t_{1/2} = 1.0$ weeks: A2–A1 gap +13.7 percentage points, McNemar $p = 6.6 \times 10^{-17}$.

- A3 recovery of the A2–A1 gap is uneven: 74% at $N = 35$, $t_{1/2} = 1.0$ weeks; 48%–57% at larger N .

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We endeavored to provide an independent, higher-resolution confirmation of the Tier 1 ranking. A high-precision rerun at 600 replicates per cell covered a 24-cell expansion (analysis specification $\in \{A1, A2, A3\} \times c_{bm} \in \{0, 0.45\} \times N \in \{35, 70\} \times t_{1/2}^{DGP} \in \{0.5, 1.0\}$ weeks, OL+BDC design), executed using 8-core ‘parallel::mclapply’. Wall-clock time was 436 seconds; convergence was 100% across all 14,400 fits. Per-replicate output is at ‘analysis/data/quick-sim/02-carryover-replicates.rds’; the Morris ADEMP cell-level summary is at ‘02-carryover-summary.txt’. Cell MC SE on power is approximately 0.020 in the middle of the range. The rerun adds independent confirmation of the Tier 1 ranking by exploiting within-replicate correlation among specifications via paired McNemar tests on the common simulated datasets.

Type I error across the twelve null cells of the rerun fell in [0.003, 0.053], so all three specifications hold nominal alpha at the explored N and DGP half-life combinations, in agreement with the Tier 1 result in the preceding section.

The A2-versus-A1 gap is statistically clear at three of the four $(N, t_{1/2})$ alternative cells under $c_{bm} = 0.45$. The binding cell is at $t_{1/2}^{DGP} = 1.0$ weeks: at $N = 35$, $\Delta_{A2-A1} = +5.7$ percentage points ($0.400 \rightarrow 0.457$), McNemar $p = 1.2 \times 10^{-4}$; at $N = 70$, the gap widens to +13.7 percentage points ($0.737 \rightarrow 0.873$), McNemar $p = 6.6 \times 10^{-17}$. The shorter-half-life cells show $\Delta_{A2-A1} \approx +2$ to +3 percentage points with McNemar $p \in [0.004, 0.015]$, a small but statistically resolvable advantage; it is reduced because at $t_{1/2}^{DGP} = 0.5$ weeks the carryover decays quickly enough that the binary on-drug A1 specification already captures most of the structure that A2 is designed to recover.

A3’s recovery of the A2-versus-A1 gap is uneven across the rerun grid. Only the small- N short-half-life cell ($N = 35$, $t_{1/2}^{DGP} = 0.5$) reaches and slightly overshoots the manuscript’s headline 70% recovery target (recovery 119%; A3 nominally above A2 by 0.5 percentage points). At $N = 35$, $t_{1/2}^{DGP} = 1.0$ weeks, recovery is 74%; the larger- N cells fall to 57% and 48%. Paired McNemar on A2 versus A3 finds A2 strictly dominant only at the high-leverage corner ($N = 70$, $t_{1/2}^{DGP} = 1.0$ weeks, $\Delta = +7.2$ percentage points, $p = 1.2 \times 10^{-7}$); in the other three cells the paired difference lies between -0.5 and $+1.5$ percentage points with McNemar $p \geq 0.18$, and the two specifications are operationally tied at this resolution. Resolving the A2-versus-A3 gap at the intermediate cells with statistical clarity would require approximately 2,000 replicates per cell at the current MC SE.

The substantive implication, carried into the Discussion, is that the summary statement ‘A2 dominates A1 and A3 captures roughly 70% of the gap’ is

correct in spirit at the prazosin-calibrated reference cell ($N = 35$, $t_{1/2}^{DGP} = 1.0$ weeks), but the A3-as-default heuristic is supported only at small N and short half-life. The production-grade recommendation in the Discussion reports the recovery percentages across the full $(N, t_{1/2}^{DGP})$ grid.

5 Discussion

5.1 The principal finding

A2 dominates A1 and A3 by approximately 10 percentage points at the reference cell, and this ranking is stable across all design and DGP variation in the Tier 1 grid.

- Reference cell: $A2 = 0.860$, $A1 = 0.766$, $A3 = 0.770$; ≈ 5 MC SEs of separation at $n_{sim} = 500$.
- Ranking $A2 > A1 \approx A3$ holds across all designs, architectures, sample sizes, and carryover half-lives.
- The stability of the ranking is itself informative: no design or carryover condition displaces A2.

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The central result of this factorial study is relatively straightforward to state. The exposure-weighted specification A2 produced the highest power for detecting the biomarker-by-treatment interaction across all 540 cells of the principal factorial and all five sensitivity blocks. The advantage over the binary-treatment specification A1 and the lagged-nuisance specification A3 was approximately 10 percentage points at the reference configuration (Architecture B, Hybrid design, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$ weeks; $A2 = 0.860$, $A1 = 0.766$, $A3 = 0.770$), corresponding to approximately five Monte Carlo standard errors of separation at $n_{sim} = 500$. The ranking $A2 > A1 \approx A3$ held across all three trial designs, both DGP architectures, both sample sizes, and all carryover half-lives in the Tier 1 grid. We note that the stability of this ranking is, in some respects, as informative as the ranking itself: it appears that neither the choice of trial design nor the magnitude of carryover is sufficient to displace A2 from its dominant position.

5.2 Robustness to decay-form variation

A2 retains its advantage across all five DGP decay forms, even when the analyst assumes exponential and the true form is linear or Weibull.

- A2 does not fall below A1 or A3 under any evaluated mis-specification condition.
- The linear DGP produces the largest A2 penalty (sharp cutoff poorly approximated by an exponential tail), yet the advantage is preserved.

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- A2 is more robust to decay-form mis-specification than anticipated.

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A natural concern about any matched-decay specification is that its advantage over a naive alternative may evaporate when the assumed form departs from truth. We have examined this directly. The heatmap in Figure reffig:fig-heatmap-mismatch crosses DGP decay form against analysis specification at the reference cell ($t_{1/2} = 1.0$ week, Architecture B, Hybrid design, $N = 70$, $c_{bm} = 0.45$). A2 retains its advantage across all five DGP decay forms (linear, exponential, Weibull at $k \in \{0.7, 1.0, 1.5\}$) when the analyst correctly specifies the form. But what happens when the analyst assumes exponential while the true DGP is not? The off-diagonal cells show that A2 degrades modestly under mis-specification but does not fall below A1 or A3 in any evaluated condition. The linear DGP produces the largest A2 penalty, because the sharp cutoff at $2t_{1/2}$ is poorly approximated by an exponential tail, yet even there the advantage is preserved. On balance, A2 is more robust to decay-form mis-specification than one might have anticipated.

5.3 Robustness to half-life mis-specification

Block S2 contradicts the Jones-Kenward expectation: A3 does not overtake A2 at any evaluated half-life mismatch, because A2's curvature is too gentle to be costly under mis-specification.

- A2 power span across a four-fold mis-specification: ≈ 0.066 ; A2 remains above A1 and A3 throughout.
- Four-fold undershoot (assumed 0.25 wks, true 1.0 wks): $A2 = 0.766$ vs. $A1/A3 \approx 0.74$.
- A3 closes the gap as half-life information degrades, but practical magnitudes are too small for A3 to overtake A2.

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Block S2 provides the strongest practical reassurance in the study. Jones and Kenward [-@jones2014crossover] developed the crossover-trial framework underlying the A3 specification with the expectation that a non-parametric carry-over indicator should outperform a parametric model under mis-specification, because the lagged indicator does not commit to a functional form. The present results do not support that expectation, at least in the regimes we

have examined. When the analyst's assumed half-life departs from the true half-life by a factor of two or four (assumed $t_{1/2} \in \{0.25, 0.5, 1.0, 2.0\}$ at true $t_{1/2} = 1.0$), A2 power ranges from 0.766 to 0.832, a span of roughly 0.066 across the full range and roughly 0.024 across the factor-of-two window (numerical values are given in Figure reffig:fig-sens-s2). A1 and A3, by construction not consuming the assumed half-life, are invariant to this mis-specification. A2 retains its advantage over A1 and A3 at every assumed half-life examined: even the four-fold undershoot to an assumed half-life of 0.25 weeks leaves A2 at 0.766, still above A1 and A3 at roughly 0.74. It would appear that the exposure-weighted predictor's curvature is sufficiently gentle that mis-specification is less consequential than the crossover-trial literature anticipated. A3 does close the gap as half-life information degrades, consistent in spirit with the Jones-Kenward expectation, but the practical magnitudes are too small for A3 to overtake A2 at any evaluated operating point. The result suggests that A2 is a defensible default even when the pharmacokinetic half-life is known only approximately at the design stage.

5.4 Autocorrelation sensitivity

Contrary to expectation, A2's 'corCAR1' structure does not extract disproportionately more information at higher ρ ; the advantage is approximately constant across $\rho \in [0.5, 0.9]$.

- A2 advantage: ≈ 6 pp at $\rho = 0.5$, ≈ 5 pp at $\rho = 0.9$.
- All three specifications gain power proportionally with ρ ; relative gains are comparable.
- Autocorrelation acts additively, not interactively, on the specification ranking.

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Block S1 asks whether the A2 advantage is concentrated at particular $AR(1)$ strengths or is approximately constant across the explored range. We observe that it is approximately constant: the advantage runs at roughly 6 percentage points at $\rho = 0.5$ and narrows to roughly 5 points at $\rho = 0.9$, a range too narrow to be practically meaningful. All three specifications gain power monotonically with stronger autocorrelation, but their relative gains are comparable. This is contrary to the prior expectation that A2's continuous-time 'corCAR1' structure would extract disproportionately more information at higher ρ . It appears, rather, that autocorrelation and carryover strength act additively on the specification ranking: stronger serial correlation lifts every specification by roughly the same amount, leaving the $A2 > A1 \approx A3$ ordering undisturbed.

5.5 Dropout and the limits of specification refinement

Dropout is far more consequential than the carryover specification choice; at 30% dropout all three specifications are near floor and effectively indistinguishable.

- At 30% MCAR, all specifications collapse to power 0.12–0.25; A2 provides no protection against the power loss itself.
- MAR at 10% is approximately equivalent to MCAR at 12%, because MAR preferentially censors high-severity participants.
- Dropout prevention is more consequential than specification refinement; trialists should prioritise accordingly.

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Block S3 is, in some ways, the most sobering result in the study. The three specifications are meaningfully distinguished at low-to-moderate dropout: at 10% MCAR, power drops from approximately 0.86 to approximately 0.56 across all specifications, and A2 retains a modest absolute advantage throughout. At 30% dropout, however, all specifications are near floor (0.12 to 0.25) and are effectively indistinguishable. Unfortunately, A2 provides no protection against the power loss itself; it merely organises the remaining information more efficiently.

Under MAR dropout biased by baseline severity, the pattern is qualitatively similar but shifted slightly: MAR dropout at 10% is comparable to MCAR dropout at approximately 12%, which is expected because MAR preferentially censors high-severity participants who contribute more to the interaction signal under Architecture B.

The practical implication is direct. Dropout prevention, whether through shorter trial durations, participant engagement strategies, or adaptive scheduling, is far more consequential for the biomarker-interaction test than the choice of analysis-model carryover specification. Trialists who invest in refining the carryover specification while tolerating high dropout rates have, in our view, the priorities somewhat reversed.

5.6 Architecture dependence

Architecture B amplifies the A2 benefit because carryover erodes the biomarker-response correlation directly, and A2 recovers more of that attenuated signal.

- Architecture B is more sensitive to carryover under all three decay forms and all Weibull shapes examined.
- The A2 advantage is larger under Architecture B than Architecture A, consistent with the correlation-erosion mechanism.
- The $A2 > A1 \approx A3$ ranking holds under both architectures.

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1181 *The companion manuscript establishes that Architecture B (MVN differential*
1182 *correlation) loses substantially more power under carryover than Architecture*
1183 *A (direct mean moderation). The present results confirm that this qualitative*
1184 *gap is robust to decay-form variation: Architecture B is more sensitive to*
1185 *carryover under all three decay forms and at all five Weibull shape values*
1186 *examined. The A2 advantage is larger under Architecture B than under Ar-*
1187 *chitecture A, which is consistent with the mechanism. Under Architecture B,*
1188 *carryover erodes the biomarker-response correlation directly, and the continu-*
1189 *ous exposure-weighted predictor recovers more of that attenuated signal than*
1190 *either the binary on-drug indicator or the lagged-nuisance term can. The A2*
1191 *> A1 ≈ A3 ranking holds under both architectures, though the magnitude*
1192 *of the benefit warrants closer attention in trials where the data-generating*
1193 *mechanism is presumed to follow Architecture A.*

1194 5.7 The latent-subtype question

1195 **The present simulation does not evaluate a finite-mixture DGP; whether**
1196 **A2 dominance extends to discrete-subtype data is an open question.**

- 1197 • A mixture DGP would produce discrete responder/non-responder profiles
1198 rather than a graded covariance structure.
- 1199 • Under a mixture DGP, A3's form-free carryover indicator might gain an
1200 advantage over A2.
- 1201 • The present results cannot address this; a mixture DGP evaluation is iden-
1202 tified as a direction for subsequent work.

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1210 *A reviewer of the companion manuscript broached the question of whether*
1211 *Architecture B's covariance-based interaction is adequately modelled by the*
1212 *MVN differential-correlation parameterisation, or whether an explicit finite-*
1213 *mixture DGP would be more faithful to the latent-subtype hypothesis. The*
1214 *question merits acknowledgment here because it bears on the present results.*
1215 *A true mixture DGP would produce biomarker-response associations that are*
1216 *qualitatively different from those generated by a continuous correlation: dis-*
1217 *crete responder and non-responder subgroups rather than a graded covariance*
1218 *structure. Under a mixture DGP, the lagged-treatment specification A3 might*

gain an advantage over A2, because A3 does not assume a specific decay form and may be more tolerant of the non-smooth response profiles that mixture data can produce. The present simulation does not evaluate a mixture DGP and cannot address this question directly. We have not been able to determine, as yet, whether the A2 dominance we observe extends to genuinely discrete-subtype data-generating processes; this is a direction that merits investigation in subsequent work.

5.8 Comparison with prior simulation studies

The present factorial replicates Hendrickson’s baseline and extends it by contrasting three analysis specifications, with the A2-versus-A1 advantage being a new contribution.

- Hendrickson (2020): power 0.82 reproduced here at 0.860 ± 0.016 ; A1/A3 comparators are the new element.
- Jones-Kenward expectation (A3 dominates under mis-specification) is not supported at the prazosin-calibrated parameters.
- Sturdevant-Lumley two-tier reporting convention is adopted and extended to the biomarker-interaction estimand.

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Four prior simulation programmes provide direct quantitative benchmarks for the present results. We discuss each in turn, focusing on cells where the prior parameter ranges overlap with ours and noting where the design targets differ.

Hendrickson [-@hendrickson2020] introduced the biomarker-by-treatment interaction question for aggregated N-of-1 trials and reported power for the matched-decay (A2) analysis under exponential carryover at $t_{1/2} = 1$ week, $N = 70$, $c_{bm} = 0.45$, Hybrid design. The reported power was 0.82 at 1,000 replicates under a single architecture. The present 500-replicate factorial reproduces this result at the same cell to 0.860 ± 0.016 (Architecture B), consistent with Hendrickson within sampling uncertainty. The principal methodological extension here is the addition of A1 and A3 as comparators; Hendrickson did not contrast analysis-side specifications, so the A2-versus-A1 advantage of approximately 10 percentage points is a new contribution rather than a replication.

Jones and Kenward [-@jones2014crossover] developed the crossover-trial framework underlying the A3 specification with the expectation, noted above, that the non-parametric lagged-on indicator should outperform a parametric carryover model under decay-form mis-specification. The present S2 sensitivity block directly tests this expectation and finds it is not supported in the regimes we have examined: although A2’s power varies by roughly 0.066 across a four-fold half-life mis-specification, A2 retains its advantage at every assumed half-life, so A3 does not become preferred at any evaluated mismatch

level. The Jones-Kenward expectation is preserved in spirit, in that A3 closes the gap as half-life information degrades, but the practical magnitudes are too small for A3 to overtake A2 in the explored window. The disagreement is informative: it appears that the exposure-weighted predictor's curvature is gentle enough that mis-specification is less costly than the crossover-trial literature anticipates, at least at the prazosin-calibrated parameters.

Senn [-@senn2016mastering] established the variance-decomposition foundation for within-subject designs, predicting that within-subject contrasts are disproportionately advantageous when between-patient variance $\sigma_{\text{between}}^2$ is large relative to within-patient variance. The present simulation does not vary $\sigma_{\text{between}}^2$ as a primary axis, but the S1 autocorrelation sensitivity block bears on the related question of how within-subject information accumulates with serial correlation. Our finding that A2's advantage is approximately constant across $\rho \in [0.5, 0.9]$ is qualitatively consistent with Senn's prediction that within-subject methods extract roughly proportional information across correlation regimes; the absence of a strong ρ interaction with the analysis specification is the new result here.

Sturdevant and Lumley [-@sturdevant2021carryover] developed the two-tier reporting convention we have adopted: a principal factorial covering the main contrast, supplemented by marginal sensitivity blocks anchored at a reference cell. Their convention was designed for simulation studies whose main claim is a ranking or contrast rather than a precise power estimate, and our adoption of it is documented in the 'simulation-grid-plan.md' design document. The two-tier structure proved useful in practice: the Tier 1 factorial establishes the $A2 > A1 \approx A3$ ranking, while the Tier 2 blocks probe its robustness to single-axis perturbations. The extension to the biomarker-interaction estimand is, to the best of our knowledge, new.

The companion manuscripts at 'analysis/report/01-dgp-mean-moderation-vs-mvn/' and 'analysis/report/05-nof1-design-sensitivity/' provide complementary benchmarks: the first establishes the architecture-dependence of carryover-induced power loss confirmed here in the architecture-dependence subsection, and the second reports the effect of carryover on the Hybrid N-of-1 design's saturation profile confirmed in §4.4.

5.9 Limitations

Five limitations bound the generalisability of the findings: single prazosin parameterisation, incomplete analysis strategy space, OFAT sensitivity blocks, omitted OL design, and monotone dropout assumption.

- Absolute power values are specific to the prazosin-PTSD parameterisation; rankings are expected to be more portable but are not verified across alternative trajectory shapes.
- Three specifications do not exhaust the carryover-aware model space; weighted analysis, random slopes, and observation exclusion are not evaluated.
- Sensitivity blocks are OFAT; joint perturbations (e.g., high ρ combined with high dropout) are not fully explored.

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Several limitations should be noted when interpreting these results.

First, the simulation draws its response-trajectory parameters from a single configuration calibrated to the Hendrickson (2020) prazosin-PTSD framework (Gompertz maxima, rates, and dispersions). The absolute power values are specific to this parameterisation. The ranking of analysis specifications is expected to be more portable, but we have not been able to verify this across alternative response profiles. Trialists working in therapeutic areas with substantially different trajectory shapes should regard the power estimates as illustrative rather than definitive.

Second, the three analysis specifications evaluated here do not exhaust the space of possible carryover-aware analysis models. Weighted analysis, within-subject contrasts, two-stage random slopes, and exclusion of contaminated observations (discussed in the companion manuscript's Section 6) are not evaluated. We have endeavored to select the three specifications that represent the most natural practical alternatives for the N-of-1 context, but a more comprehensive comparison would cross the present grid with these additional strategies, and others may merit consideration.

Third, the sensitivity blocks are one-factor-at-a-time perturbations around a single reference configuration. Unfortunately, joint perturbations, for example high ρ combined with high dropout, may produce interactions that the marginal blocks do not capture. The exploratory block S5 ($\rho \times$ carryover) provides one such joint check and does not reveal a qualitative interaction, which is reassuring, but the space of possible joint combinations is large and our exploration of it is necessarily incomplete.

Fourth, the Tier 1 grid omits the open-label (OL) design because OL lacks the off-drug contrast required by the A1 and A3 specifications. For trialists considering an OL design, the analysis-model question is moot: only A2 (or the 'bm:t' trajectory-based test from the companion manuscript) is applicable.

Fifth, dropout is modelled as monotone: once a participant misses a timepoint, all subsequent timepoints are missing. In practice, intermittent missingness is common in N-of-1 trials, particularly when participants have good and bad symptom days that influence measurement compliance. Intermittent missingness would presumably be less destructive than monotone dropout because it preserves some late-trial observations that contribute to the treatment contrast. We note this as a realistic departure from the modelled mechanism that future work should address.

The headline Tier 1 ranking is further supported by the high-precision rerun reported at the end of the Results section, which resolves the A2-versus-A1 gap to MC SE 0.020 with paired McNemar $p < 10^{-3}$ at the headline cells.

6 Conclusions

1. **A2 (exposure-weighted) is the recommended default.** Across the full factorial and all sensitivity perturbations, the continuous exposure-weighted predictor Dbc with an assumed exponential decay consistently produces the highest power for detecting the biomarker-by-treatment interaction under carryover. The advantage over the binary-treatment (A1) and lagged- nuisance (A3) alternatives is approximately 10 percentage points at the reference configuration and is robust to DGP decay-form variation, within-factor autocorrelation, and analyst-truth half-life mismatch.
2. **Half-life mis-specification does not displace A2.** A2 power varies by roughly 0.066 across a four-fold mis-specification of the analyst's assumed half-life (Block S2), but A2 retains its advantage over A1 and A3 at every assumed half-life examined. Trialists who have only an approximate pharmacokinetic estimate can adopt A2 with confidence that a mis-specified half-life does not cost A2 its ranking.
3. **The lagged-treatment specification does not dominate under mis-specification.** Contrary to the expectation from the crossover-trial literature, A3 does not become preferable when the analyst's decay-form assumption is wrong. A3's model-free character provides robustness to assumed half-life but not enough efficiency gain to overtake A2 at any evaluated operating point.
4. **Dropout is more consequential than the carryover specification.** Power degrades sharply with dropout rate under all specifications. At 20% dropout, all three specifications lose approximately half their power relative to the no-dropout condition. Trialists should prioritise dropout prevention over carryover-specification refinement.
5. **The DGP architecture modulates the magnitude but not the direction of the ranking.** Architecture B (MVN differential correlation) amplifies the benefit of A2 because carryover-induced correlation erosion makes the continuous predictor's information advantage more consequential. Under Architecture A (direct mean moderation), A2's advantage is smaller but still positive.
6. **Reporting recommendations.** Simulation studies evaluating biomarker-moderated treatment effects under carryover should report: (a) the analysis-model carryover specification used,
 - (b) the assumed decay form and half-life, (c) sensitivity of power estimates to at least one alternative specification, and (d) the anticipated dropout rate, which is the dominant source of power loss in the evaluated grid.
7. **Type-I error is well-controlled.** All three specifications maintain nominal or conservative type-I error rates across the full Tier 1 grid and the S4 effect-size curve, with no evidence of inflation under any DGP decay form,

1397 architecture, or carryover level.

1398 7 Conflict of interest

1399 The authors declare no conflicts of interest.

1400 8 Funding

1401 This work received no specific funding.

1402 9 Data availability statement

1403 **All simulation outputs and drivers are versioned in the pmsimstats-ng**
1404 **repository; exact paths are given in the Reproducibility section.**

- 1405 • Per-replicate Monte Carlo outputs and cell-level summaries: `analysis/data/`.
- 1406 • Simulation drivers: `analysis/scripts/carryover-sensitivity/`.
- 1407 • Exact file paths listed in the Reproducibility section below.

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1415 *The data and code supporting the findings of this study are available in*
1416 *the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and*
1417 *ADEMP-compliant cell-level summaries are versioned under ‘analysis/data/’;*
1418 *simulation drivers are at ‘analysis/scripts/’. Exact paths for this manuscript*
1419 *are listed in the Reproducibility section below.*

1420 10 Reproducibility

1421 **The manuscript is fully reproducible from pinned package versions in**
1422 **renv.lock, deterministic seeds, and pre-computed checkpoints; no sim-**
1423 **ulation code re-executes during knit.**

- 1424 • Environment: R 4.4.x, zzcollab Docker container; packages pinned in
1425 `renv.lock`.
- 1426 • Principal outputs: `01-factorial.rds` and `04-sensitivity.rds`; cell-
1427 level summaries at `analysis/data/02-grid-summary.rds`.
- 1428 • Manuscript Rmd loads pre-computed checkpoints only; no simulation code
1429 runs during render.

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This manuscript was rendered with R 4.4.x inside the project's zcollab Docker container (image hash recorded in 'Dockerfile'; package set captured in 'renv.lock'). Principal packages used by the simulation pipeline are 'nlme' (continuous-time linear-mixed analysis with 'corCAR1' residual correlation), 'parallel' (8-core 'mclapply' for parameter-grid sweeps), 'data.table' (the 'original-extended' simulation collection), 'furry' and 'purrr' (the 'tidyverse' simulation collection), and 'rmarkdown' plus 'bookdown' for the manuscript build.

***Random-number generator.** Each simulation driver sets 'set.seed(20260507)' at the top of the replicate loop and uses 'parallel::mc.set.seed' inside 'mclapply'. Per-replicate seeds are derived from the master seed as `master + rep_idx`.*

***Data and code availability.** Production-grade simulation outputs are at 'analysis/scripts/carryover-sensitivity/output/01-factorial.rds' (principal factorial) and '04-sensitivity.rds' (sensitivity blocks S1 through S5). Cell-level summaries are at 'analysis/data/02-grid-summary.rds' and 'analysis/data/02-sensitivity-summary.rds'. Driver scripts are '01-run-factorial.R', '04-run-sensitivity-blocks.R', and '02-summarise-grid.R' under 'analysis/scripts/carryover-sensitivity/'. The simulation grid plan and design rationale are documented at 'analysis/scripts/carryover-sensitivity/simulation-grid-plan.md', which serves as the Morris ADEMP pre-registration. High-precision prototype rerun outputs are at 'analysis/data/quick-sim/02-carryover-replicates.rds' (per-replicate) and '02-carryover-summary.txt' (Morris ADEMP cell-level summary). The manuscript source, paper-specific bibliography ('references.bib'), and SIM CSL file are at 'analysis/report/02-carryover-sensitivity/'.*

11 References

1. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*. 2019;38(11):2074-2102. doi:10.1002/sim.8086